



STEAMSPACE

Pre-Field-Trip Character Planner

6th grade · complete in your classroom before the field trip · ~20–30 min

Teacher guide — read before handing out

1. **Name the theme** — Tell students the class theme (desert animal, story hero, water-cycle mascot, simple-machines robot, etc.). Tie it to what they're studying this unit.
2. **Original only** — No franchise or copyrighted characters. Students invent their own.
3. **Two pages, no tech** — Students need only a pencil. Page 1 = draw the character. Page 2 = rebuild it with shapes.
4. **Bring it on field-trip day** — Students bring both pages to STEAMSPACE. Coaches use the shape plan to jump straight into Tinkercad.

My character

My name	Date	Class theme (from teacher)

My character's name: _____

My character connects to our theme because: _____

Draw your character below

Draw your original character here

Look at your drawing — what shapes do you see?

Body part	What shape does it look like?

Reconfigure with shapes

Now rebuild your character using **3D shapes** — just like you will in Tinkercad! These are the building blocks you'll use on field-trip day.

Shape menu (Tinkercad primitives)

Shape	Looks like	Good for
Box	 box	body, legs, feet, base
Cylinder	 tube	arms, legs, hats, towers
Sphere	 ball	head, eyes, balls
Cone	 cone	hats, noses, spikes
Torus (donut)	 ring	rings, belts, handles
Wedge	 wedge	ramps, shoulders, angles
Roof	 roof	pointed tops, hats, roofs

Break your character into shapes

Body part	Tinkercad shape	Size (big / medium / small)

Re-draw your character using only those shapes

Boxes, cylinders, spheres, cones — it doesn't need to be perfect. You're planning your build!

Draw your character with shapes here

The biggest shape is: _____ (*build this one first in Tinkercad*)

The smallest shape is: _____

One thing I want to figure out on field-trip day: _____

Bring both pages with you to STEAMSPACE!

When you arrive, your coach will look at your shape plan and help you start building in Tinkercad right away. Students who arrive with a completed planner get a head start on the design block.